UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Maxima and Minima of a Function of Two Variables

Complete study pack for class 009: local maxima, minima and saddle points, the necessary and sufficient conditions, the test with r, s, t and $rt - s^2$, and the working rule

Compiled from the PESU class 009 slide deck and AV-summary transcript, the Unit 1 class-work sheet and past ESA papers. Generated 15 October 2026.

· How to use this pack

Class 009 is the theory session: what a relative maximum, minimum and saddle point are, and the test that decides between them. The ESA problems (stationary points of a cubic, applied problems) are worked in the class 010 pack, and constrained problems by Lagrange's method in class 011. The worked examples here are short ones that show each branch of the test, including the doubtful case $rt - s^2 = 0$. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S009·7	Deck 009, slide 7 (rendered .pptx)
T009 00:07:04	AV-summary transcript 009 (“Extremas of functions”), segment start
CW U1 P31	Unit 1 class-work problem 31
ESA Dec-23 1c	Past paper: Dec 2023, question 1.c

Contents

1. 1 · Maxima, minima and saddle points
2. 2 · Definitions: relative maximum, minimum and extreme value
3. 3 · Conditions and the working rule
4. Appendix A · Symbol glossary
5. Appendix B · Sources, coverage and gaps

1 Maxima, minima and saddle points

A surface can have several peaks (local maxima) and several pits (local minima); at each the tangent plane is horizontal. A saddle point also has a horizontal tangent plane, but the surface rises in one direction and falls in another, like a horse saddle. The word "relative" (or local) means the comparison is only with nearby points.

Key terms and definitions

Saddle point. A stationary point at which $f(a,b)$ is neither a maximum nor a minimum: the surface rises in some directions and falls in others. Test: $rt - s^2 < 0$. S009·5 · S009·7 · T009 00:11:06

Graphs and diagrams

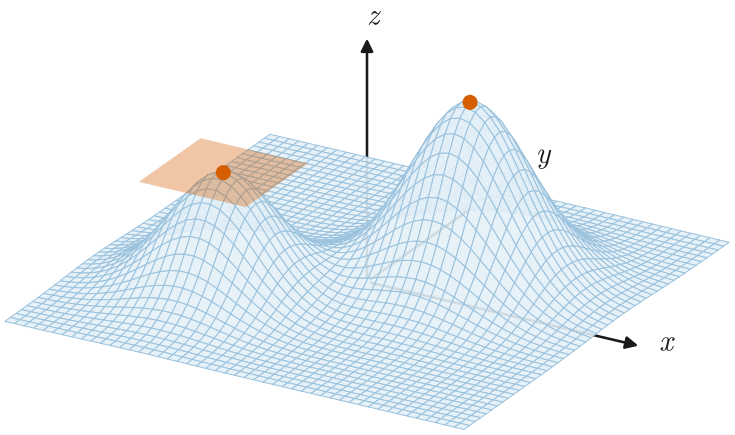


Figure G1.1. Local maxima: peaks with horizontal tangent planes. ESA ★

S009·3 · T009 00:00:16

How to read it. A surface with two peaks (illustrative: a sum of two bell shapes). Both peaks are local maxima; the taller one is also the largest value overall. The vermilion square is the horizontal tangent plane at the lower peak.

$z = x^2 + y^2$

minimum at $(0,0)$: $rt - s^2 = 4 > 0,\ r = 2 > 0$

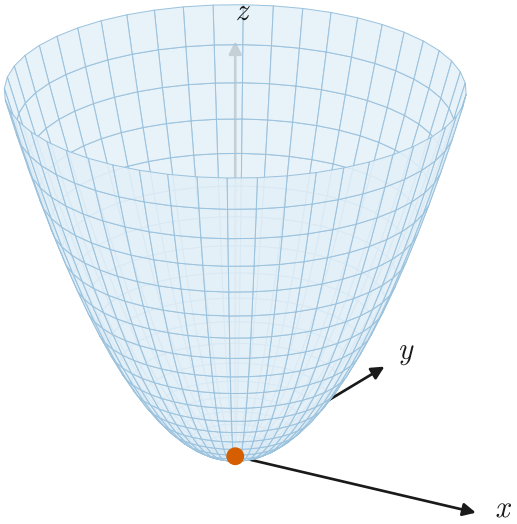


Figure G1.2. A local minimum: $z = x^2 + y^2$. ESA ★

S009·4 · T009 00:02:11

How to read it. The bowl $z = x^2 + y^2$ has its lowest point at the origin. Every nearby point is higher: $\Delta > 0$. Its test values are $r = t = 2$, $s = 0$, $rt - s^2 = 4 > 0$, $r > 0$ (W3.1).

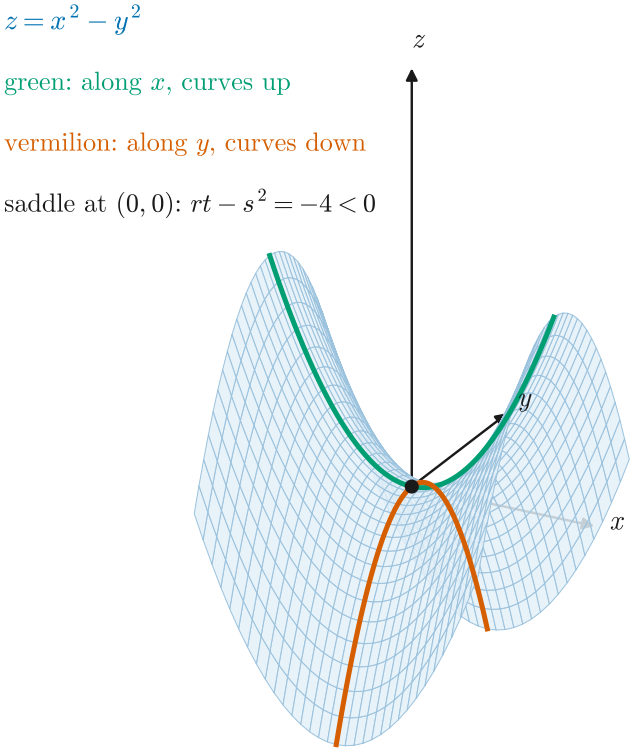


Figure G1.3. A saddle point: $z = x^2 - y^2$. ESA ★★

S009·5 · S009·7

How to read it. Along the x-direction (green) the surface curves up, along the y-direction (vermilion) it curves down. The origin is stationary but neither a maximum nor a minimum; $rt - s^2 = -4 < 0$ (W3.2).

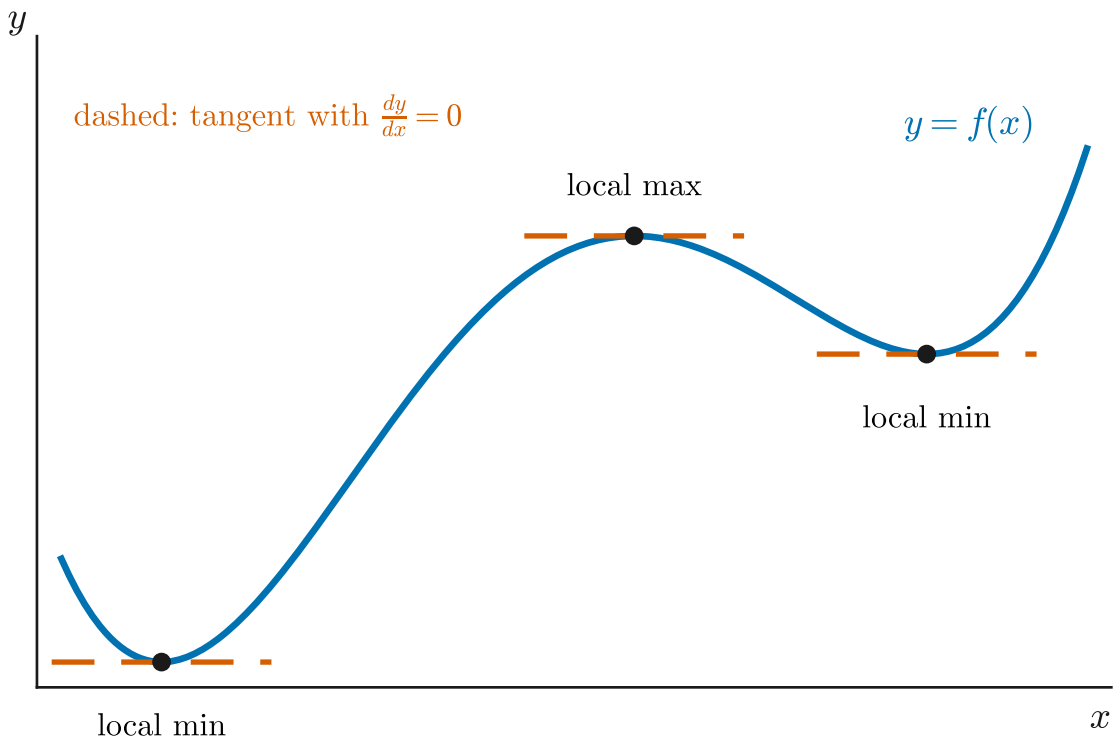


Figure G1.4. One-variable picture: maxima, minima and $\frac{dy}{dx} = 0$. ESA ★

T009 00:00:50 · T009 00:01:10

How to read it. On $y = f(x)$ the tangent is horizontal (slope 0) at every local maximum and minimum (short dashed segments). The two-variable test copies this: $f_x = f_y = 0$ replaces $\frac{dy}{dx} = 0$, and $rt - s^2$ with r replaces $\frac{d^2y}{dx^2}$.

Important points and why

1. One variable recap: solve $\frac{dy}{dx} = 0$ for critical points; $\frac{d^2y}{dx^2} < 0$ gives a maximum, > 0 a minimum; $= 0$ gives no conclusion. The tangent at a maximum or minimum is parallel to the x-axis (slope 0). ESA ★★ T009 00:00:50 · T009 00:01:40 · T009 00:02:57
2. In two variables the tangent plane at a maximum or minimum is horizontal (parallel to the xy-plane); slide 009·3 shows a horizontal tangent plane at a relative maximum (Figure G1.4). ESA ★ S009·3 · S009·4 · T009 00:01:10
3. A function can have several local maxima and several local minima (slides 009·3–4: "These are all local maxima / minima"). ESA ★ S009·3 · S009·4 · T009 00:00:16

Exam trap. In one variable $\frac{d^2y}{dx^2} = 0$ is not "a saddle point"; it is only inconclusive (x^4 has a minimum, x^3 an inflection). ADDED FOR CLARITY T009 00:03:14

SHORT-NOTE RECAP

Local maxima are peaks, local minima are pits; at both the tangent plane is horizontal. A saddle point also has a horizontal tangent plane but rises one way and falls another. A function can have several of each.

2 Definitions: relative maximum, minimum and extreme value

f has a relative maximum at (a, b) if $f(a, b) > f(a + h, b + k)$ for all small h, k, and a relative minimum if $f(a, b) < f(a + h, b + k)$. Equivalently, $\Delta = f(a + h, b + k) - f(a, b)$ keeps one sign: negative for a maximum, positive for a minimum. Either is an extreme value.

Key terms and definitions

Relative maximum. A function $f(x, y)$ is said to have a relative maximum at a point (a, b) if there exists a neighbourhood of the point (a, b), say $(a + h, b + k)$, h and k small, such that $f(a, b) > f(a + h, b + k)$. S009·6 · T009 00:03:46

Relative minimum. If $f(a, b) < f(a + h, b + k)$ then $f(x, y)$ is said to have a relative minimum at (a, b). S009·6 · T009 00:04:58

Extreme value. $f(a, b)$ is said to be an extreme value of f if it is either a maximum or a minimum. S009·6

Neighbourhood. The points very close to (a, b): $(a + h, b + k)$ with h, k small. "Relative" means largest or smallest only in such a neighbourhood, not over the whole plane. T009 00:04:12 · T009 00:04:40

> *"In other words if $\Delta = f(a + h, b + k) - f(a, b)$ is of the same sign for all small values of h, k and if this sign is negative then $f(a, b)$ is maximum & if this sign is positive then $f(a, b)$ is a minimum."* S009·6 · T009 00:05:23

Formulas

E2.1 · Sign of the increment ESA ★

$$\Delta = f(a + h, b + k) - f(a, b) \qquad \begin{cases} < 0 \text{ for all small } h, k: & \text{maximum} \\ > 0 \text{ for all small } h, k: & \text{minimum} \end{cases}$$

gives	The definition of an extreme value.
applies	Used to settle the doubtful case $rt - s^2 = 0$.
note	Same sign in every direction.

S009·6 · T009 00:05:23

Important points and why

1. "Relative" because only points in a neighbourhood of (a, b) are compared, not the whole plane. ESA ★ T009 00:04:40

SHORT-NOTE RECAP

Relative maximum at (a, b): $f(a, b) > f(a + h, b + k)$ for small h, k; minimum: $<$. Equivalently $\Delta = f(a + h, b + k) - f(a, b)$ has one sign: negative ⇛ maximum, positive ⇛ minimum. Either is an extreme value.

3 Conditions and the working rule

As in one variable, first make the first derivatives zero: $f_x = 0$, $f_y = 0$ give the stationary (critical) points. Then use the second derivatives $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$ and the Hessian determinant $rt - s^2$: $rt - s^2 > 0$ with $r < 0$ is a maximum, with $r > 0$ a minimum; $rt - s^2 < 0$ is a saddle point; $rt - s^2 = 0$ is doubtful and needs further investigation.

Key terms and definitions

Stationary (critical) point. A point where $f_x = 0$ and $f_y = 0$: the tangent plane is horizontal, and the function may have an extreme value there. S009·8 · T009 00:02:47 · T009 00:06:19

Hessian. The determinant $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = rt - s^2$ of second partial derivatives. S009·7 · T009 00:07:22

> *"The necessary and sufficient condition for $f(x, y)$ to have a maximum value at (a, b) is $f_x(a, b) = 0$, $f_y(a, b) = 0$ and $rt - s^2 > 0$, where $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$. $rt - s^2$ is the determinant of the Hessian matrix $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix}$."* S009·7

> *"Note: If $rt - s^2 < 0$, $f(a, b)$ is not an extreme value, i.e. $f(x, y)$ is neither a maximum nor a minimum at the point (a, b) and the point is called a saddle point. If $rt - s^2 = 0$, then the test provides no information about the nature of the point."* S009·7

Formulas

E3.1 · Stationary points (necessary condition) ESA ★★★

$$f_x = \frac{\partial f}{\partial x} = 0, \qquad f_y = \frac{\partial f}{\partial y} = 0$$

gives	The candidates $(a, b), (c, d), \ldots$ for extreme values.
applies	f differentiable; solve the two equations simultaneously.
note	Two-variable version of $\frac{dy}{dx} = 0$.

S009·7 · S009·8 · T009 00:06:19

E3.2 · Second derivatives used in the test ESA ★★★

$$r = \frac{\partial^2 f}{\partial x^2}, \qquad s = \frac{\partial^2 f}{\partial x\, \partial y}, \qquad t = \frac{\partial^2 f}{\partial y^2}$$

gives	The three numbers at each stationary point.
applies	Evaluate at each stationary point separately.
note	$s = f_{xy} = f_{yx}$ (Clairaut).

S009·7 · S009·8 · T009 00:07:04

E3.3 · Hessian determinant

ESA ★★★

$$rt - s^2 = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - f_{xy}^2$$

GIVES Whether the stationary point is an extremum.

APPLIES At a stationary point.

NOTE Positive: extremum (decide by r). Negative: saddle. Zero: doubtful.

S009·7 · T009 00:07:37

E3.4 · The test (working rule, steps 3–6)

ESA ★★★

$$rt - s^2 > 0,\ r < 0 \ \Rightarrow\ \text{maximum}; \qquad rt - s^2 > 0,\ r > 0 \ \Rightarrow\ \text{minimum}$$
$$rt - s^2 < 0 \ \Rightarrow\ \text{saddle point}; \qquad rt - s^2 = 0 \ \Rightarrow\ \text{doubtful}$$

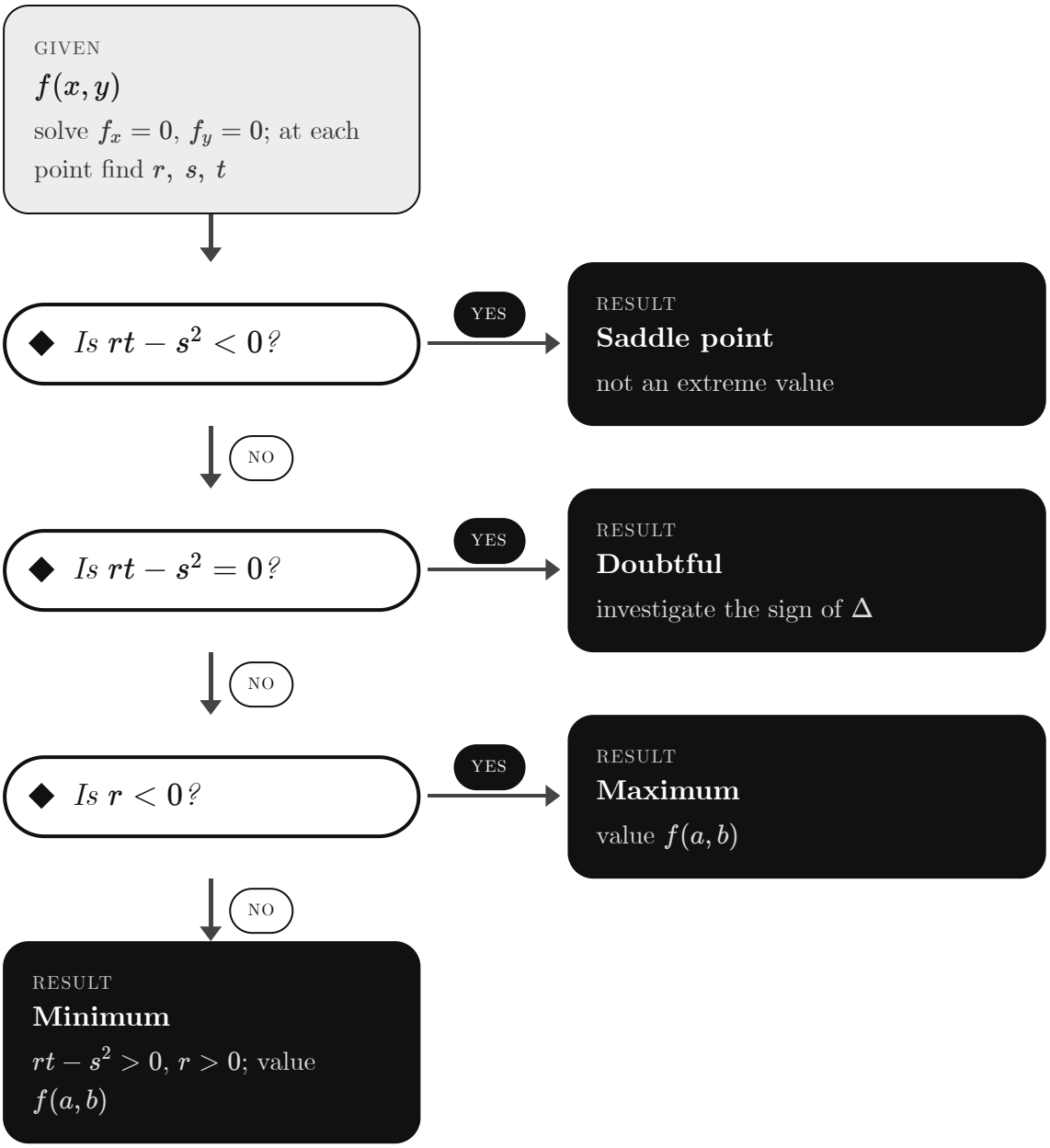
GIVES The nature of each stationary point.

APPLIES After E3.1–E3.3.

NOTE $r < 0$ plays the role of $\frac{d^2y}{dx^2} < 0$ in one variable. When $rt - s^2 > 0$, r and t have the same sign, so t can be used instead.

S009·8 · S009·9 · T009 00:11:32

Flowcharts



Remark. Repeat the ladder for every stationary point.

Exam trap. Never classify from r alone.

Figure F3.1. Working rule: nature of a stationary point ESA ★★★

S009·8 · S009·9 · T009 00:11:32

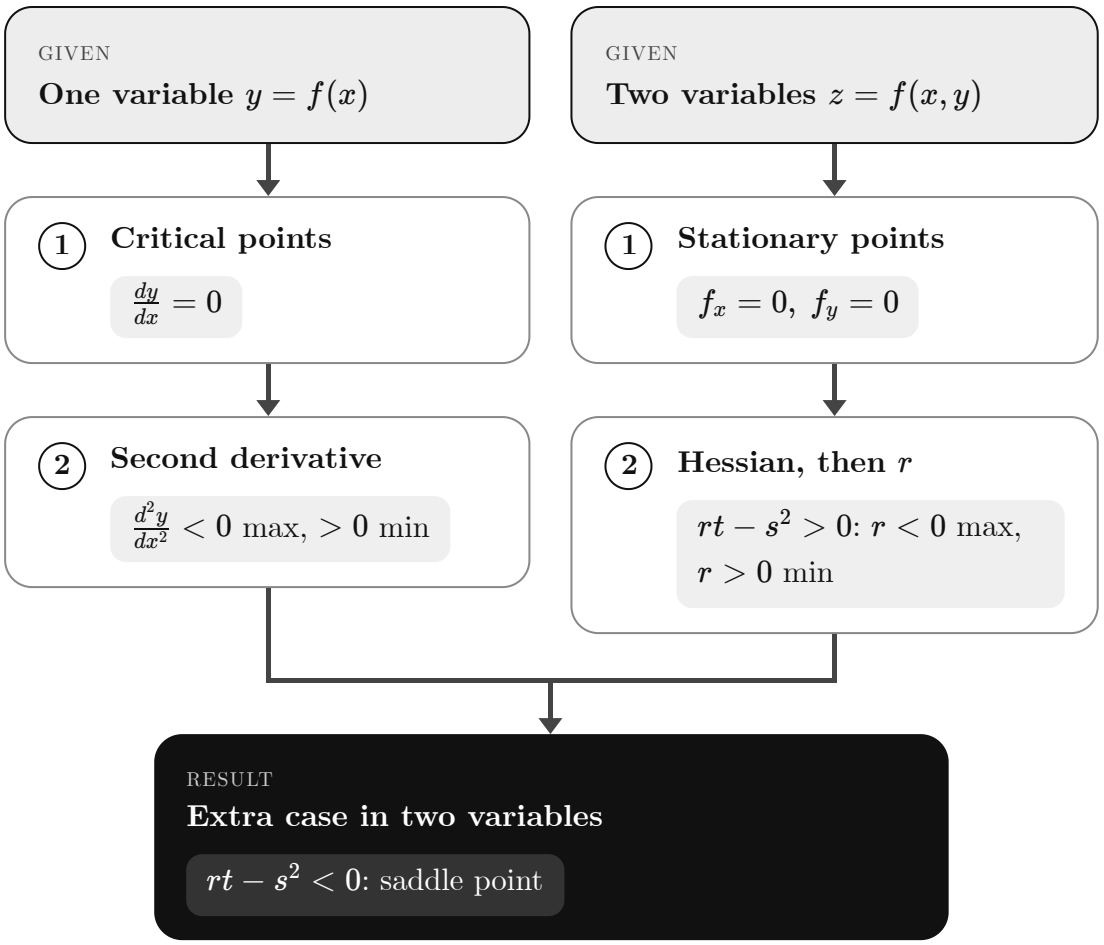


Figure F3.2. From one variable to two ESA ★★

T009 00:06:00 · T009 00:10:20

The four outcomes of the test

$rt - s^2$	r	Nature of (a, b)	Example at the origin
> 0	< 0	Maximum	$-x^2 - y^2$
> 0	> 0	Minimum	$x^2 + y^2$ (W3.1)
< 0	any	Saddle point	$x^2 - y^2$ (W3.2)
$= 0$	any	Doubtful: further investigation	$x^4 + y^4$ (min), $x^4 - y^4$ (neither)

S009·7 · S009·8 · S009·9

Important points and why

1. Working rule step 1: find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, equate each to zero, solve simultaneously; let $(a, b), (c, d), \ldots$ be the pairs of values. ESA ★★★ S009·8 · T009 00:11:32
2. Step 2: calculate $r = \frac{\partial^2 f}{\partial x^2}$, $s = \frac{\partial^2 f}{\partial x \partial y}$, $t = \frac{\partial^2 f}{\partial y^2}$ for each pair of values. ESA ★★★ S009·8

3. Steps 3–6: $rt - s^2 > 0$ and $r < 0$: maximum; $rt - s^2 > 0$ and $r > 0$: minimum; $rt - s^2 < 0$: not an extreme value, (a, b) is a saddle point; $rt - s^2 = 0$: the case is doubtful and needs further investigation. ESA ★★★ S009·8 · S009·9 · T009 00:12:33
4. Two important conditions to remember (lecture): both first partials zero, and $rt - s^2$ from the second partials. ESA ★★★ TRANSCRIPT ONLY T009 00:08:50
5. Unlike one variable, the sign of r alone is not enough: first check $rt - s^2 > 0$, then look at r. ESA ★★★ T009 00:07:51 · T009 00:09:43
6. The value of the function at a maximum or minimum point is the maximum or minimum value: always substitute the point back into f. ESA ★★ ADDED FOR CLARITY T009 00:04:51
7. Each stationary point is tested separately: a function can have maxima, minima and saddle points together (class 010: four points for $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$). ESA ★★★ ESA Dec-23 1c · ESA Jan-24 s1 1a

Why must f_x and f_y both vanish at an extremum?
Along the line $y = b$, $f(x, b)$ is a function of x alone with an extremum at $x = a$, so its derivative $f_x(a, b) = 0$. The same along $x = a$ gives $f_y(a, b) = 0$. Geometrically, the tangent plane is horizontal.

ADDED FOR CLARITY T009 00:06:19

Why does $rt - s^2 < 0$ mean a saddle?
Near a stationary point $f(a + h, b + k) - f(a, b) \approx \frac{1}{2}(rh^2 + 2shk + tk^2)$ (Taylor, class 007). This quadratic keeps one sign for all h, k exactly when $rt - s^2 > 0$ (sign of r); when $rt - s^2 < 0$ it is positive in some directions and negative in others.

ADDED FOR CLARITY S009·6 · S009·7

Why is r enough to separate maximum from minimum when $rt - s^2 > 0$?
$rt > s^2 \geq 0$, so r and t have the same sign: the surface curves the same way along x and along y.

ADDED FOR CLARITY S009·8

Exam trap. $rt - s^2 < 0$ is a saddle point (a definite answer); only $rt - s^2 = 0$ is "no conclusion". The lecture mixes the two at one point. S009·7 · T009 00:08:23 · T009 00:11:06

Exam trap. Do not decide maximum or minimum from r before checking $rt - s^2 > 0$: $x^2 - 3xy + y^2$ has $r = 2 > 0$ at the origin but $rt - s^2 = -5$: a saddle. ADDED FOR CLARITY S009·8

Exam trap. $f_x = f_y = 0$ is necessary, $rt - s^2 > 0$ sufficient; slide 009·7 calls the pair "necessary and sufficient". ADDED FOR CLARITY S009·7

Exam trap. Evaluate r, s, t at each stationary point; a sign that holds at one point need not hold at another. S009·8

Worked examples

W3.1 · A bowl: minimum ESA ★

GIVEN $f(x,y) = x^2 + y^2$.

STEP 1 $f_x = 2x = 0$, $f_y = 2y = 0$: stationary point $(0,0)$.

STEP 2 $r = 2$, $s = 0$, $t = 2$; $rt - s^2 = 4 > 0$ and $r > 0$.

ANSWER Minimum at $(0,0)$; minimum value 0

CHECK $f \geq 0$ everywhere, so the answer agrees with the definition.

ADDED FOR CLARITY S009·4 · S009·8

W3.2 · A saddle ESA ★

GIVEN $f(x,y) = x^2 - y^2$.

STEP 1 $f_x = 2x = 0$, $f_y = -2y = 0$: $(0,0)$.

STEP 2 $r = 2$, $s = 0$, $t = -2$; $rt - s^2 = -4 < 0$.

ANSWER $(0,0)$ is a saddle point

CHECK Along the x-axis $f = x^2 \geq 0$, along the y-axis $f = -y^2 \leq 0$: Δ changes sign.

ADDED FOR CLARITY S009·5 · S009·9

W3.3 · Doubtful case resolved: a minimum ESA ★

GIVEN $f(x,y) = x^4 + y^4$.

STEP 1 $f_x = 4x^3$, $f_y = 4y^3$: stationary point $(0,0)$.

STEP 2 $r = 12x^2 = 0$, $s = 0$, $t = 12y^2 = 0$: $rt - s^2 = 0$, doubtful.

STEP 3 Definition: $\Delta = f(h,k) - f(0,0) = h^4 + k^4 > 0$ for all small $(h,k) \neq (0,0)$.

ANSWER Minimum at $(0,0)$ (value 0)

CHECK The test gave no information; the sign of Δ settles it.

ADDED FOR CLARITY S009·9 · S009·6

W3.4 · Doubtful case resolved: neither ESA ★

GIVEN $f(x, y) = x^4 - y^4$.

STEP 1 Stationary point $(0, 0)$; again $r = s = t = 0$, so $rt - s^2 = 0$.

STEP 2 $\Delta = h^4 - k^4$: positive for $k = 0$, negative for $h = 0$.

ANSWER Neither a maximum nor a minimum at $(0, 0)$

CHECK Compare with W3.3: $rt - s^2 = 0$ can hide either answer.

ADDED FOR CLARITY S009·9 · S009·6

W3.5 · The working rule on a quadratic ESA ★★

GIVEN Find the maximum and minimum values of $f(x, y) = 4x^2 + 2y^2 + 4xy - 10x - 2y - 3$.

STEP 1 $f_x = 8x + 4y - 10 = 0$, $f_y = 4y + 4x - 2 = 0$.

STEP 2 Subtract: $4x - 8 = 0$, so $x = 2$, $y = -\frac{3}{2}$: one stationary point $(2, -\frac{3}{2})$.

STEP 3 $r = 8$, $s = 4$, $t = 4$: $rt - s^2 = 32 - 16 = 16 > 0$, $r > 0$.

STEP 4 $f(2, -\frac{3}{2}) = 16 + \frac{9}{2} - 12 - 20 + 3 - 3 = -\frac{23}{2}$.

ANSWER Minimum value $-\dfrac{23}{2}$ at $\left(2, -\frac{3}{2}\right)$; no maximum

CHECK CW answer agrees; sympy agrees.

CW U1 P31

SHORT-NOTE RECAP

Solve $f_x = 0$, $f_y = 0$. At each point find $r = f_{xx}$, $s = f_{xy}$, $t = f_{yy}$. $rt - s^2 > 0$: $r < 0$ max, $r > 0$ min; $rt - s^2 < 0$: saddle; $rt - s^2 = 0$: doubtful (use Δ). Substitute the point in f for the value.

A Symbol glossary

Symbol	Name	Meaning	Appears in
(a, b)	Stationary point	Solution of $f_x = f_y = 0$	E3.1
$h,\ k$	Small increments	Neighbouring point $(a+h, b+k)$	E2.1
Δ	Increment of f	$f(a+h, b+k) - f(a, b)$	E2.1
r	f_{xx}	Second partial in x at the point	E3.2–E3.4
s	f_{xy}	Mixed partial at the point (not arc length)	E3.2–E3.4
t	f_{yy}	Second partial in y (not time)	E3.2–E3.4
$rt - s^2$	Hessian determinant	$f_{xx} f_{yy} - f_{xy}^2$	E3.3

B Sources, coverage and gaps

Source	Path	Use
Deck 009 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.*	10 slides; content on 3–9
Transcript 009 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.*	13.2 min lecture
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Problem 31 (with answer)
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	3 questions use this test (worked in class 010)
Video 009 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_*.mp4	Not opened: the slides are readable and the board work is read out

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024. The definitions are never asked by themselves, but the test of this session is the whole of three 7-mark Q1 parts: Dec 2023 Q1.c and Jan 2024 set 1 Q1.a (the same function $x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$) and Jan 2024 set 2 Q1.a ($x^3 - 12x + y^3 + 3y^2 - 9y$). They are worked in the class 010 pack; the method is starred ★★★ here.

Known gaps, conflicts and corrections

1. Slide 009·7 calls "$f_x = 0$, $f_y = 0$ and $rt - s^2 > 0$" the necessary and sufficient condition. Precisely, $f_x = f_y = 0$ is necessary (for a differentiable f) and $rt - s^2 > 0$ is sufficient; neither alone is both. The slide's "$A > 0$" is a garbled "> 0". Corrected for clarity.
2. The lecture says that in one variable a point with $\frac{d^2y}{dx^2} = 0$ is "called a saddle point" (T009 00:03:14). It is only an inconclusive case (it may be a maximum, minimum or point of inflection: x^4, $-x^4$, x^3). Corrected as a trap.
3. The lecture first says that if $rt - s^2 < 0$ "you cannot conclude anything" (T009 00:08:23) and later that it is a saddle point (T009 00:11:06). The slides (009·7, 009·9) are right: $rt - s^2 < 0 \Rightarrow$ saddle point; only $rt - s^2 = 0$ gives no conclusion.
4. The lecture's working rule at T009 00:09:26 says "if this is zero, how do I decide ... maximum or minimum"; it means $rt - s^2 > 0$. Read with the slides, the rule is: > 0 and $r < 0$ maximum, > 0 and $r > 0$ minimum, < 0 saddle, $= 0$ doubtful.
5. The lecture says mixed partials are "always equal" (T009 00:07:22); they are equal when they are continuous (Clairaut, class 001).
6. The slides give no worked example; the transcript ends with "we'll solve the problem". Problems are in class 010; the short examples W3.1–W3.4 (including the doubtful case $x^4 \pm y^4$) are added for clarity, and W3.5 is CW P31.

7. The deck's 3-D pictures are screenshots with no stated functions; Figures G1.1–G1.3 use illustrative functions with the same shapes.
8. Transcript recognition errors were corrected silently: “Hesian” → Hessian, “R T minus a s square” → $rt - s^2$, “sin” → sign, “t square y by dx square” → $\frac{d^2y}{dx^2}$, “f x” (for f_{xx}) → f_{xx}, “eight” → (a, b), “sway” → sway.

Coverage

Source	Citations	Slides cited (slide × citations)
S009	50	3×3, 4×4, 5×3, 6×8, 7×12, 8×13, 9×7
CW	1	
ESA	2	
T009	37	

Slides 009·1, 009·2 and 009·10 are title, session and closing slides. The transcript covers all content slides (3–9).